Exploratory Data Analysis on Restaurant Tips Dataset

Introduction

This project performs Exploratory Data Analysis (EDA) on a restaurant tipping dataset. The goal is to understand patterns in customer tipping behavior using statistical summaries and visualizations.

Problem Statement

The objective of this project is to perform Exploratory Data Analysis (EDA) on a restaurant tipping dataset to understand patterns in tipping behavior and identify factors that influence tips.

Data Source

The dataset is based on the classic "Tips Dataset" commonly used for data analysis and visualization practice.

Dataset Description

The dataset contains information about restaurant bills and tips.

Features in the dataset:

- total_bill : Total bill amount
- tip : Tip given by the customer
- sex : Gender of the customer
- smoker : Whether the customer is a smoker
- day : Day of the visit
- time : Lunch or Dinner
- size : Number of people in the group

Import libraries

In [2]:
```
# Data handling
import pandas as pd
import numpy as np
```

```python
# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Display settings
sns.set(style="whitegrid")
```

Load dataset

In [4]:
```python
df = pd.read_csv(r"C:\Users\moham\OneDrive\Desktop\hackathon\EDA\data/cleaned_tips.

df.head()
```

Out[4]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Dataset overview

In [5]:
```python
df.shape
```

Out[5]: (234, 7)

In [6]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 234 entries, 0 to 233
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  234 non-null    float64
 1   tip         234 non-null    float64
 2   sex         234 non-null    object
 3   smoker      234 non-null    object
 4   day         234 non-null    object
 5   time        234 non-null    object
 6   size        234 non-null    int64
dtypes: float64(2), int64(1), object(4)
memory usage: 12.9+ KB
```

In [7]:
```python
df.describe()
```

Out[7]:

	total_bill	tip	size
count	234.000000	234.000000	234.000000
mean	18.823462	2.905342	2.525641
std	7.444686	1.226852	0.913293
min	3.070000	1.000000	1.000000
25%	13.272500	2.000000	2.000000
50%	17.465000	2.750000	2.000000
75%	23.152500	3.500000	3.000000
max	40.170000	7.580000	6.000000

The above explains:

number of rows number of columns data types summary statistics

Example explanation:

The dataset contains 234 observations and 7 variables. Numerical features include total_bill, tip, and size.

check missing values

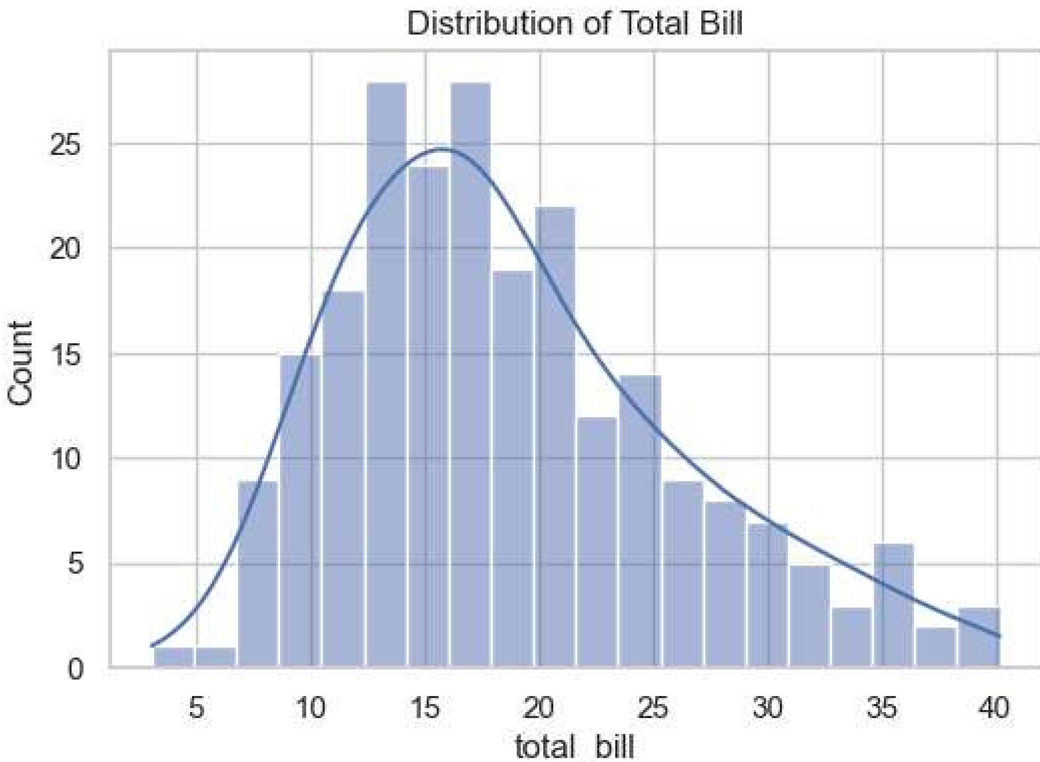
In [8]:
```
df.isnull().sum()
```

Out[8]:
```
total_bill    0
tip           0
sex           0
smoker        0
day           0
time          0
size          0
dtype: int64
```

Univariate analysis(Single variable)

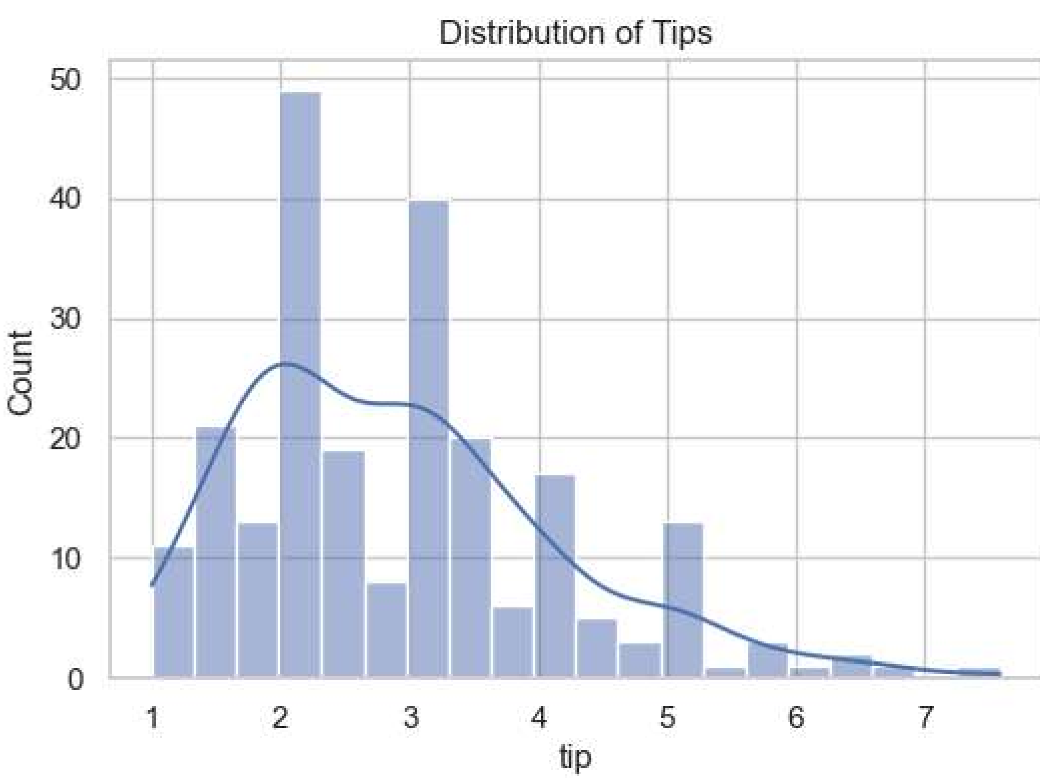
Distribution of total bill

In [9]:
```
plt.figure(figsize=(6,4))
sns.histplot(df['total_bill'], bins=20, kde=True)
plt.title("Distribution of Total Bill")
plt.show()
```



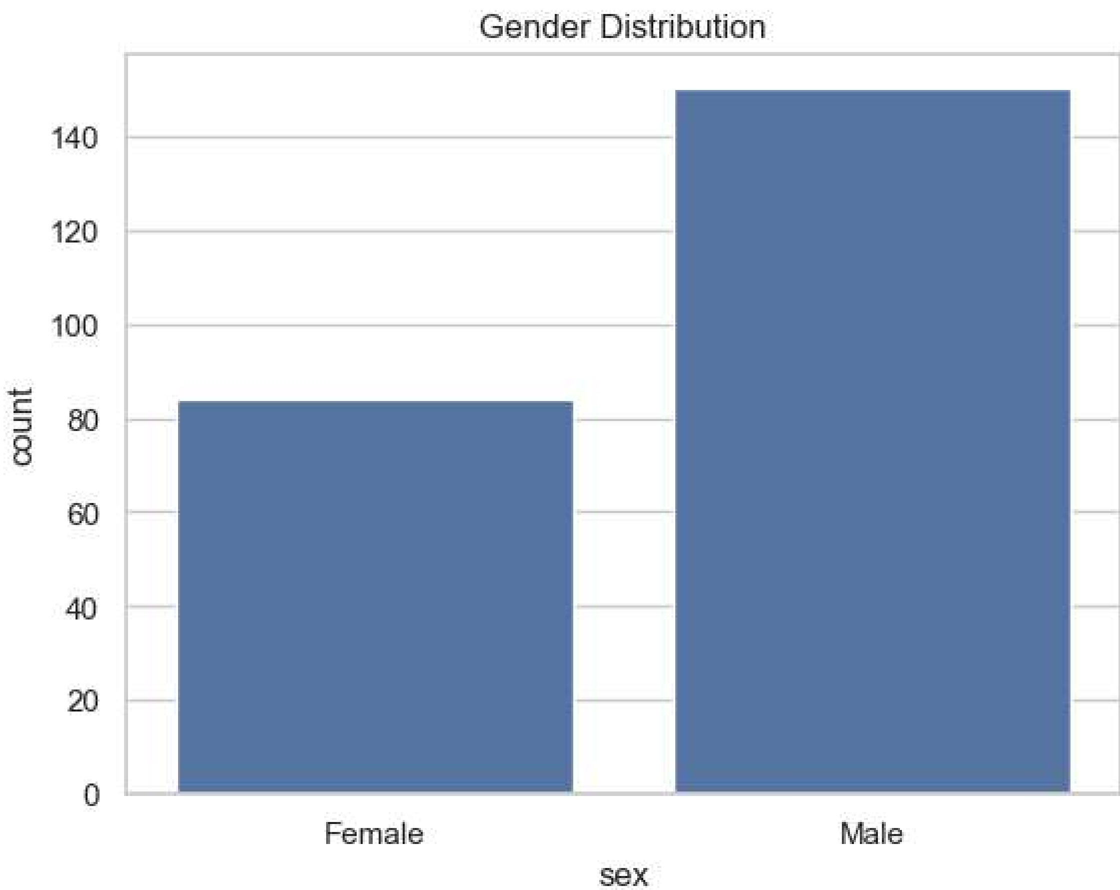
Distribution of tips

In [10]:
```
plt.figure(figsize=(6,4))
sns.histplot(df['tip'], bins=20, kde=True)
plt.title("Distribution of Tips")
plt.show()
```



Gender distribution

In [11]:
```
sns.countplot(x="sex", data=df)
plt.title("Gender Distribution")
plt.show()
```



Bivariate Analysis(Relationships)

Total bill vs Tip

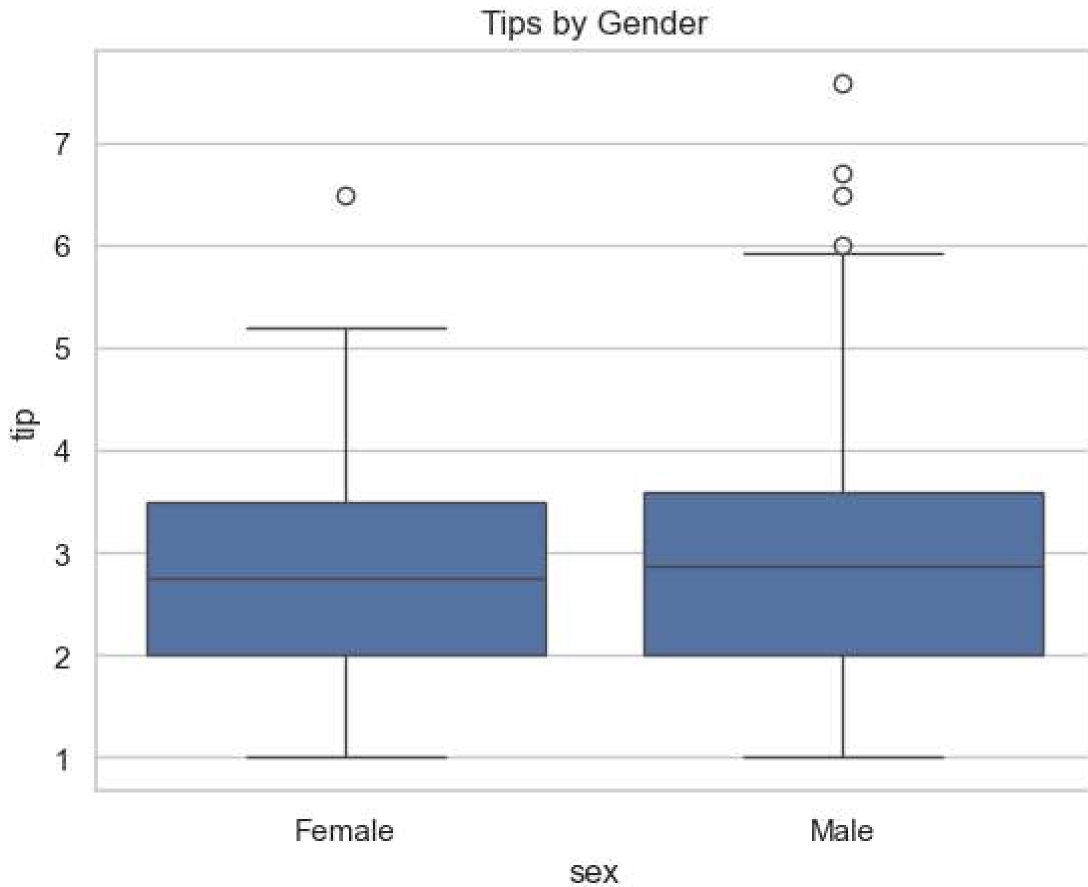
sns.scatterplot(x="total_bill", y="tip", data=df) plt.title("Total Bill vs Tip") plt.show()

Insight:

Tip amount increases with total bill, indicating a positive correlation.

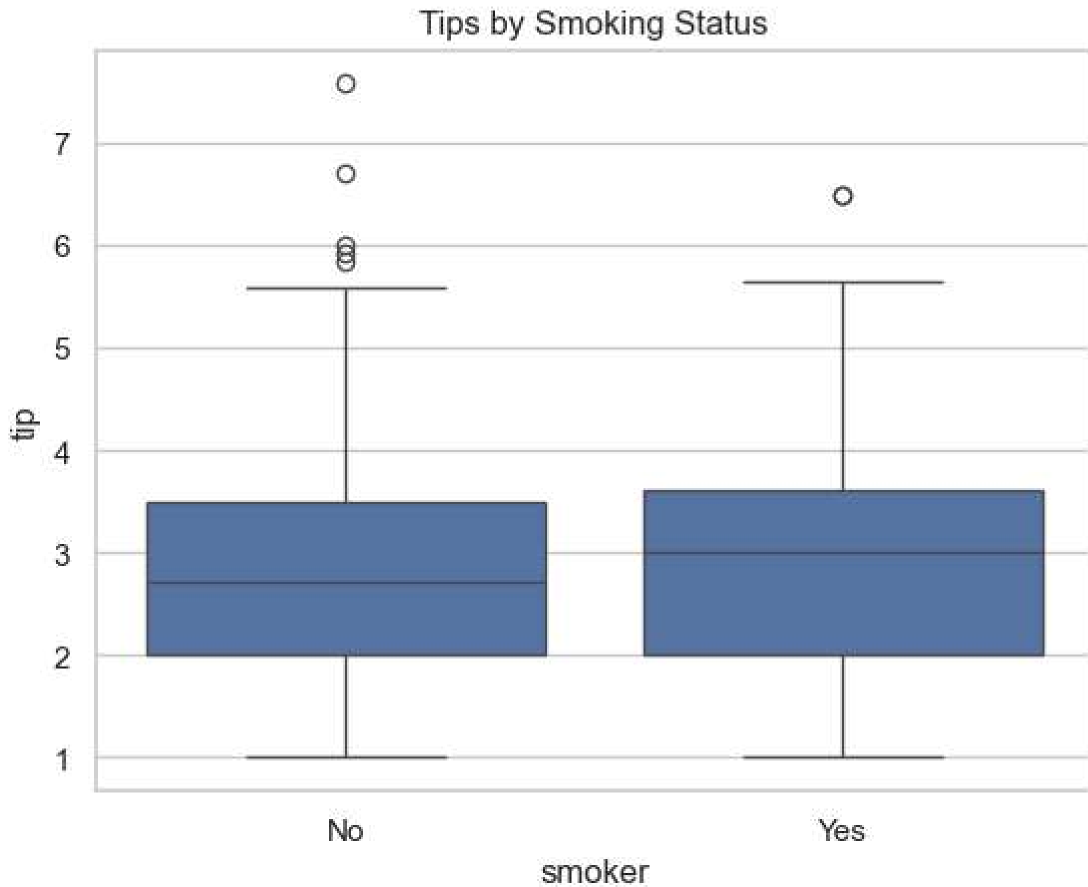
Tips by Gender

In [12]:
```
sns.boxplot(x="sex", y="tip", data=df)
plt.title("Tips by Gender")
plt.show()
```



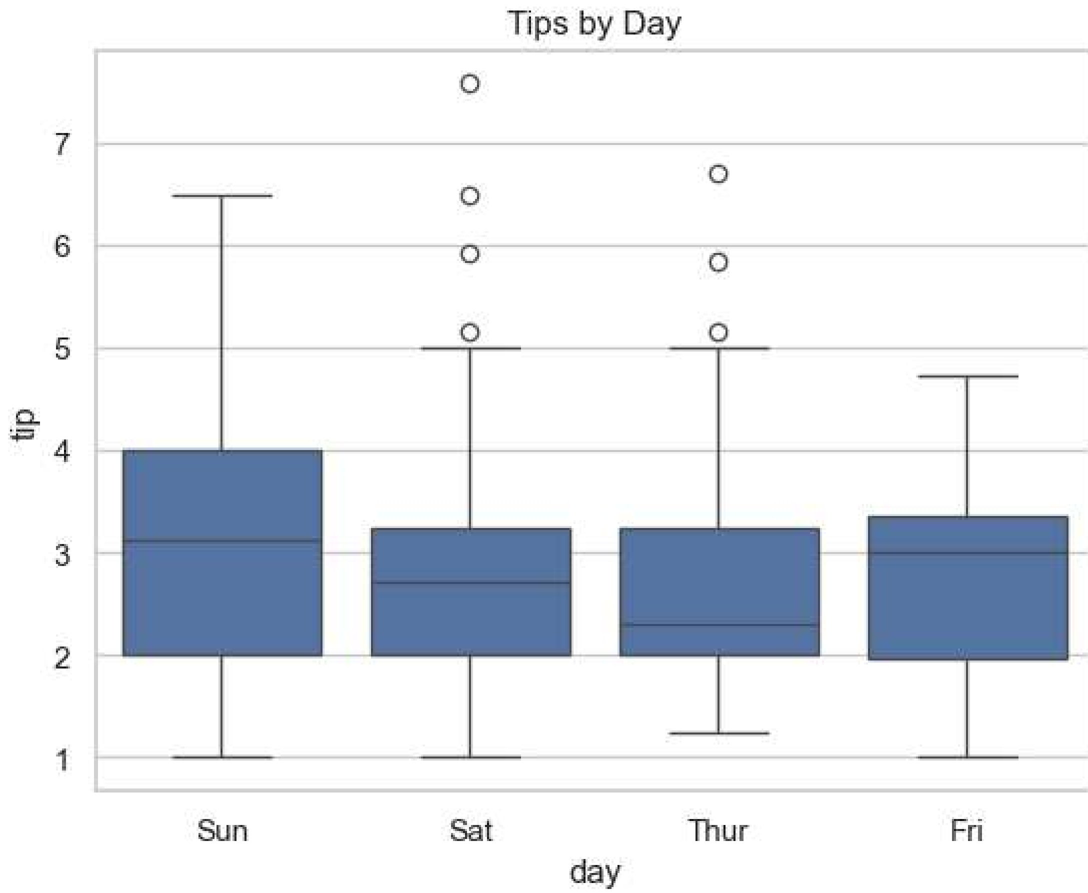
Tips by smoking status

In [13]:
```
sns.boxplot(x="smoker", y="tip", data=df)
plt.title("Tips by Smoking Status")
plt.show()
```



Tips by Day

In [14]:
```
sns.boxplot(x="day", y="tip", data=df)
plt.title("Tips by Day")
plt.show()
```

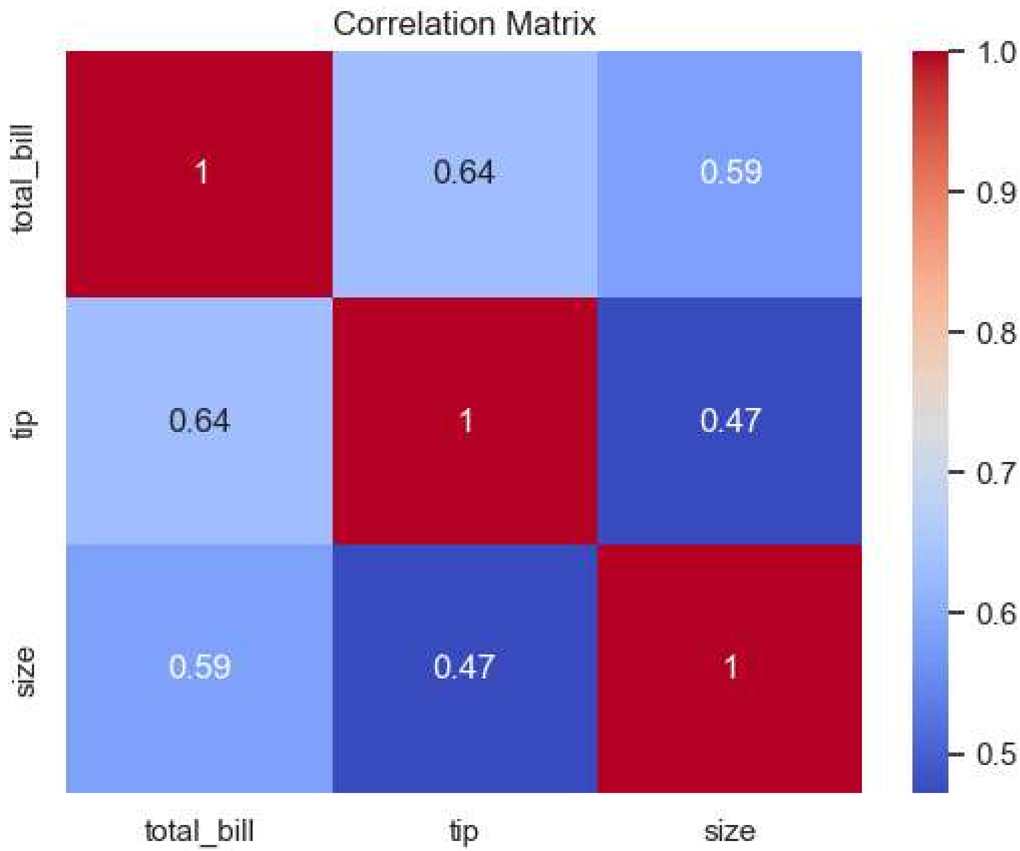


Correlation Analysis

In [15]:
```
# Convert to categorical values
numeric_df = df[['total_bill','tip','size']]
```

In [16]:
```
corr = numeric_df.corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```

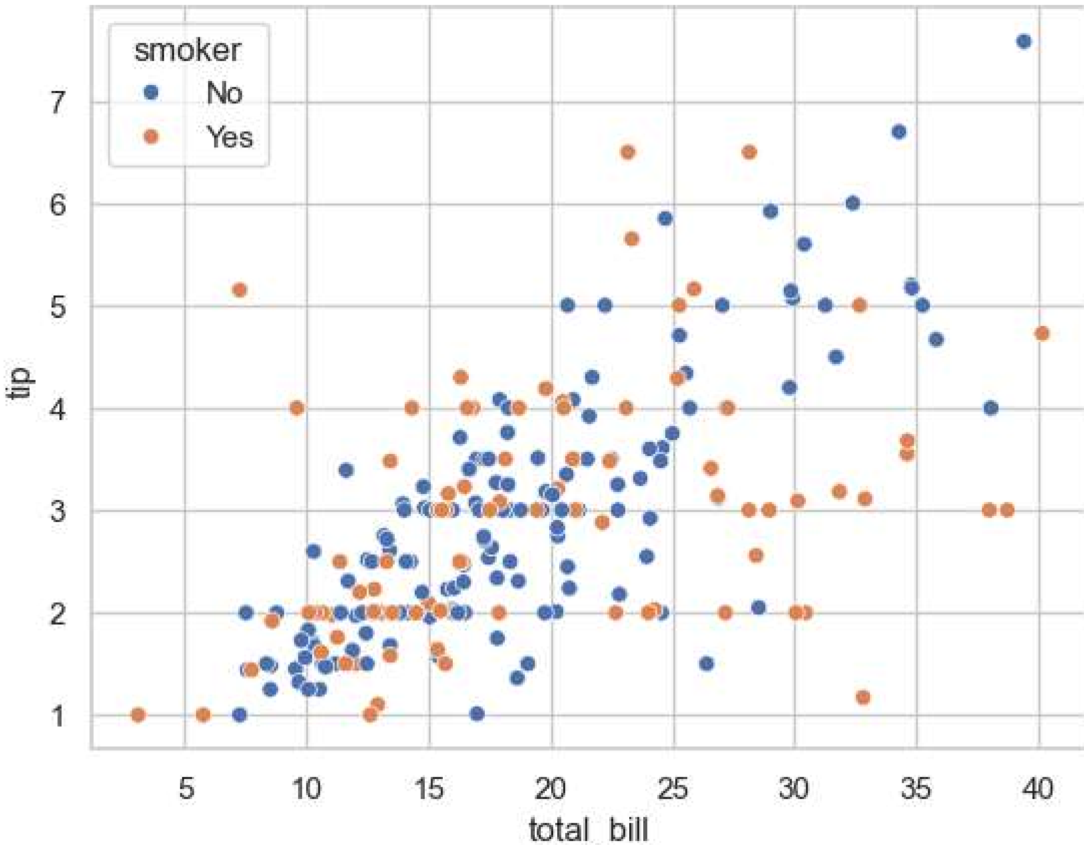


Insight:

Total bill and tip show a strong positive correlation.

Multivariate Analysis

In [17]:
```
sns.scatterplot(x="total_bill", y="tip", hue="smoker", data=df)
plt.show()
```



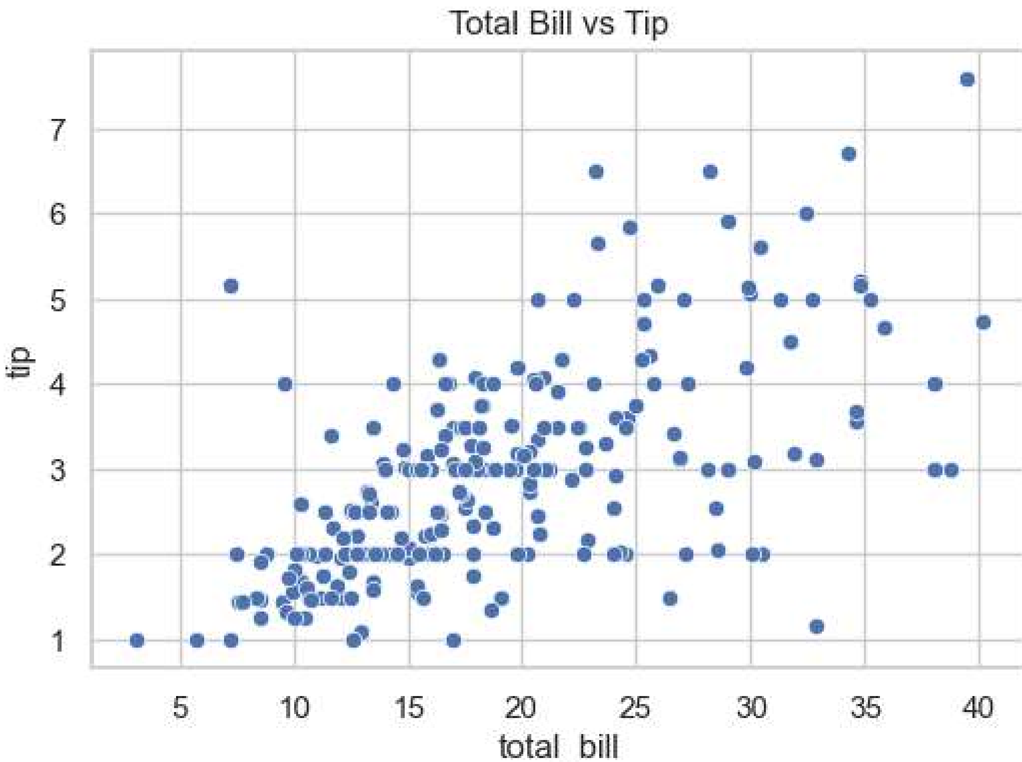
Key Findings

```
1.Total bill and tip amount have a strong positive correlation.
2.Larger groups tend to generate larger total bills.
3.Dinner time shows higher bills compared to lunch.
4.Weekend days tend to have higher tipping patterns.
```

Save Visualization

In [19]:
```
plt.figure(figsize=(6,4))
sns.scatterplot(x="total_bill", y="tip", data=df)
plt.title("Total Bill vs Tip")

plt.savefig(r"C:\Users\moham\OneDrive\Desktop\hackathon\EDA\outputs/bill_vs_tip.png
plt.show()
```



Conclusion

The analysis shows that tipping behavior strongly depends on the total bill. Larger bills tend to generate higher tips. Certain days and dinner times also show higher tipping patterns. These insights can help restaurants understand customer behavior better.